

# EBENAZER K

## Machine Learning Engineer | AI Engineer | Data Scientist (Fresher)

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### PROFESSIONAL SUMMARY

Detail-oriented Machine Learning Engineer with hands-on experience building and deploying 4+ end-to-end ML, NLP, and Generative AI applications using Python, Scikit-learn, LangChain, and FAISS. Skilled in data preprocessing, feature engineering, model evaluation, and RAG-based LLM systems, backed by a 5-month internship applying SQL and Power BI to deliver data-driven insights. Seeking a Machine Learning Engineer / AI Engineer / Data Scientist role to build scalable, production-ready AI solutions.

### TECHNICAL SKILLS

**Programming & Querying:** Python, SQL

**Data & Analysis:** NumPy, Pandas, Matplotlib, EDA, Excel

**ML Frameworks & Deployment:** Scikit-learn, Streamlit, FastAPI, Docker

**Visualization & BI:** Matplotlib, Plotly

**Version Control & Tools:** Git, GitHub, Jupyter Notebook

### MACHINE LEARNING SKILLS

**Core ML:** Data Preprocessing, Feature Engineering, Model Evaluation, Cross Validation, Hyperparameter Tuning

**Algorithms:** Linear Regression, Logistic Regression, KNN, Decision Tree, Random Forest, Naive Bayes, SVM

**NLP:** TF-IDF, NLTK, Text Classification

**Generative AI:** FAISS (Vector Search), Retrieval-Augmented Generation (RAG), LLM Applications

### PROJECTS

#### Heart Disease Prediction — End-to-End ML Application

*Technologies: Python, Scikit-learn, Streamlit, Pandas*

- Engineered and deployed an end-to-end machine learning application using Scikit-learn and Streamlit, enabling real-time heart disease risk prediction.
- Performed data preprocessing and feature scaling across multiple clinical parameters to improve model input quality and consistency.
- Evaluated model performance using Accuracy, Precision, Recall, F1-score, and Confusion Matrix, ensuring reliable classification results before deployment.

#### Customer Churn Prediction — Classification Model

*Technologies: Python, Scikit-learn, Streamlit, Pandas*

- Developed a customer churn prediction model using multiple classification algorithms to identify at-risk customers.
- Performed feature engineering, categorical encoding, and missing value handling to increase model accuracy on production-style data.
- Built an interactive Streamlit application to serve real-time churn predictions to non-technical stakeholders.

#### SMS Spam Detection — NLP Text Classification

*Technologies: Python, Scikit-learn, NLTK, Pandas, TF-IDF*

- Built an NLP text classification pipeline using tokenization, stopword removal, and TF-IDF vectorization on raw SMS data.
- Trained and compared Naive Bayes, Logistic Regression, and SVM classifiers, evaluating results via Accuracy, F1-score, and Confusion Matrix to minimize false positives.

#### AI-Powered PDF Financial Analysis Chatbot (RAG)

*Technologies: Python, LangChain, FAISS, Streamlit, Plotly*

- Built a Retrieval-Augmented Generation (RAG) application using LangChain and FAISS, implementing document chunking, vector embeddings, and semantic search to ground LLM responses in source PDFs.
- Designed an interactive Streamlit interface with Plotly visualizations to present extracted financial metrics and support natural-language PDF question answering.

#### Loan Approval Prediction — Decision Tree Classification

*Technologies: Python, Scikit-learn, Decision Tree, Joblib, Streamlit, Pandas*

- Developed a Loan Approval Prediction web app using Decision Tree Classifier; handled missing values with Mean/Mode imputation & LabelEncoder.
- Serialized trained model using Joblib and deployed an interactive web application on Streamlit Community Cloud for real-time predictions.

### INTERNSHIP

#### Data Analyst Intern — Z3 Connect, Nagercoil

5 Months

- Cleaned and preprocessed raw datasets using Python and Pandas, applying SQL queries to extract and prepare data for exploratory data analysis and feature selection.

- Built interactive Power BI dashboards and KPI reports to communicate data-driven insights to business stakeholders, supporting machine learning workflow decisions.

## EDUCATION

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### Bachelor of Technology — Artificial Intelligence and Data Science

2022 – 2026

St. Xavier's Catholic College of Engineering, Nagercoil

## CERTIFICATIONS

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- **Data Science Certification** — Networkz System
- **Python Certification** — Gelbero Pvt Limited

## GITHUB & LINKEDIN

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**GitHub:** [github.com/ebenazer890](https://github.com/ebenazer890) — ML, NLP & RAG project source code | **LinkedIn:** [linkedin.com/in/ebenazer890](https://linkedin.com/in/ebenazer890)